BIOCHEMISTRY CUME
APRIL 16, 2004

A newly discovered protein has been found to be associated with the photosynthetic apparatus associated with the inner membrane (thylakoid) of the chloroplast. The gene for this protein has been cloned and sequenced and an amino acid sequence for the protein determined by translation of the gene sequence. Analysis of this amino acid sequence using hydropathy plots suggests the presence of two transmembrane helices in the protein.

a. (10 Points) Describe two features of transmembrane helices which make them an energetically favorable way for a protein to span the membrane.

b. (20 points) What are hydropathy plots and how are they used to predict regions of proteins likely to be transmembrane helices? Answer in sufficient detail to demonstrate that you understand the theoretical basis for the plots, what is being plotted, and how they are interpreted.

c. (10 points) Based on analysis of the hydropathy plot, two orientations are possible for the protein in the membrane. Sketch the two possible orientations the protein might have in the membrane.

d. (60 points) Describe in as much detail as time and knowledge allow two experimental approaches which might be used to determine which of the two orientations is correct. In developing your experimental approaches, keep in mind that the thylakoid membrane contains over 100 different proteins in addition to this newly discovered protein, many of which are present in larger amounts. You may assume that purification procedures for the protein exist and that sufficient protein can be obtained for routine biochemical analyses. You may also assume that intact thylakoid membranes may be obtained without difficulty and that the intact membrane system may be disrupted by an osmotic shock.